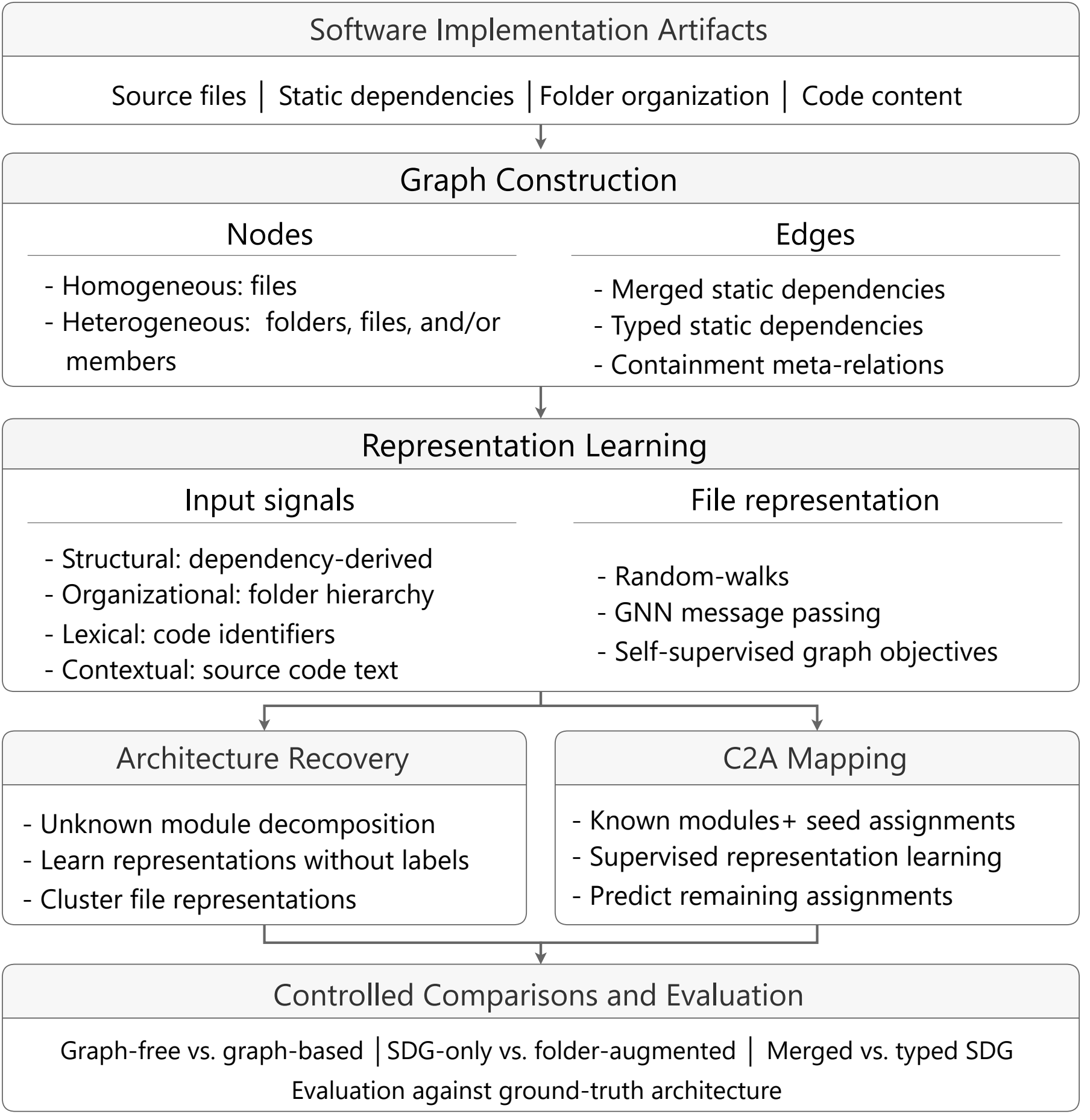
Software Implementation Artifacts
Source files | Static dependencies | Folder organization | Code content
Graph Construction
Nodes
- Homogeneous: files
- Heterogeneous: folders, files, and/or members
Edges
- Merged static dependencies
- Typed static dependencies
- Containment meta-relations
Representation Learning
Input signals
- Structural: dependency-derived
- Organizational: folder hierarchy
- Lexical: code identifiers
- Contextual: source code text
File representation
- Random-walks
- GNN message passing
- Self-supervised graph objectives
Architecture Recovery
- Unknown module decomposition
- Learn representations without labels
- Cluster file representations
C2A Mapping
- Known modules+ seed assignments
- Supervised representation learning
- Predict remaining assignments
Controlled Comparisons and Evaluation
Graph-free vs. graph-based | SDG-only vs. folder-augmented | Merged vs. typed SDG
Evaluation against ground-truth architecture